Bash Script Lab 1
System Administration Alexandria

Name: Ahmed Gamal Mohamed

Q1. Create a script that asks for user name then send a greeting to him.

Answer:

root@worker:/usr/local/bin — vim greeting

```bash
#!/bin/bash
#A script to ask for user name then send for him greeting

read  -p "Enter your user name " username
echo "Hello $username"
~
```

Verification:

root@worker:/usr/local/bin

```
[root@worker bin]# greeting
Enter your user name jimmy
Hello jimmy
[root@worker bin]#
```

Q2. Create a script called s1 that calls another script s2 where:

a. In s1 there is a variable called x, it's value 5

b. Try to print the value of x in s2 by two different ways.

Answer:

root@worker:/usr/local/bin — vim s1

```bash
#!/bin/bash
x_method1=5
x_method2=5
export x_method1
./s2 $x_method2
~
~
```

root@worker:/usr/local/bin — vim s2

```bash
#!/bin/bash

echo "This is the value of x with export method is $x_method1"

echo "This is the value of x with predefining var method is $1"
```

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Verification:

```
root@worker:/usr/local/bin
[root@worker bin]# s1
This is the value of x with export method is 5
This is the value of x with predefining var method is 5
[root@worker bin]#
```

Q3. Create a script called mycp where:

a. It copies a file to another

b. It copies multiple files to a directory.

Answer:

```
root@worker:~ — vim /usr/local/bin/mycp
#!/bin/bash

dest=${!#}
while [ "$#" -gt 1 ]; do

    cp "$1" "$dest"
    shift
done
echo "Copy complete"
~
```

Verification:

```
root@worker:~/Downloads
[root@worker ~]# ls
addscripts.sh    Desktop            f1.sh  f5.sh      output1     Templates
anaconda-ks.cfg  Documents          f2.sh  f7.sh      Pictures    Videos
backup           Downloads          f3.sh  input.txt  Public
bash.xlsx        ExamQuestions.pdf  f4.sh  Music      sendmail.sh
[root@worker ~]# mycp f* Do
Documents/ Downloads/
[root@worker ~]# mycp f* Downloads/
Copy complete
[root@worker ~]# cd Downloads/
[root@worker Downloads]# ls
f1.sh  f2.sh  f3.sh  f4.sh  f5.sh  f7.sh
[root@worker Downloads]# mycp f1.sh f1_after.sh
Copy complete
[root@worker Downloads]# cat f1.sh
hello
[root@worker Downloads]# cat f1_after.sh
hello
[root@worker Downloads]# ls
f1_after.sh  f1.sh  f2.sh  f3.sh  f4.sh  f5.sh  f7.sh
[root@worker Downloads]#
```

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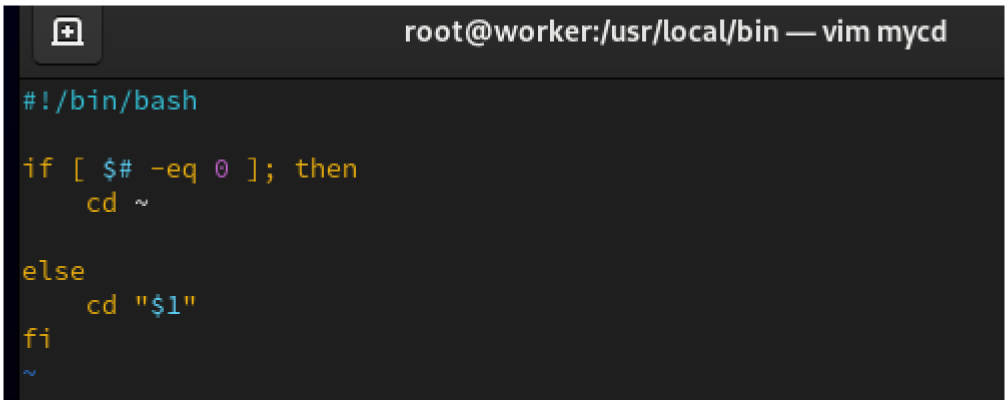
Q4. Create a script called mycd where:

a. It changed directory to the user home directory, if it is called without arguments.

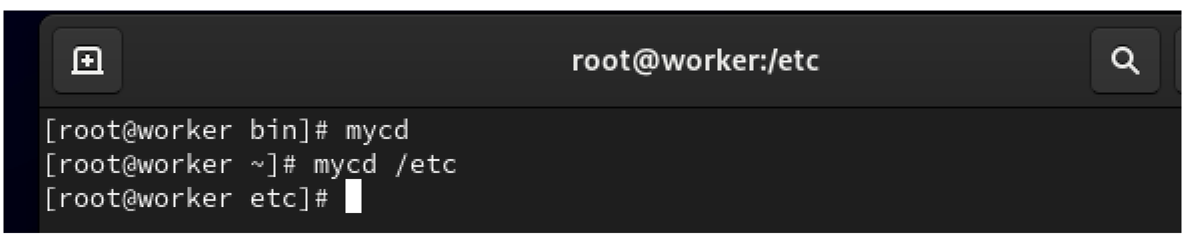
b. Otherwise, it change directory to the given directory.

Answer:

As I cant run script manipulate the directory so I puts this script as a function in bashrc to be able to run it without any tackles



Verification:



Q5. Create a script called myls where:

a. It lists the current directory, if it is called without arguments.

b. Otherwise, it lists the given directory.

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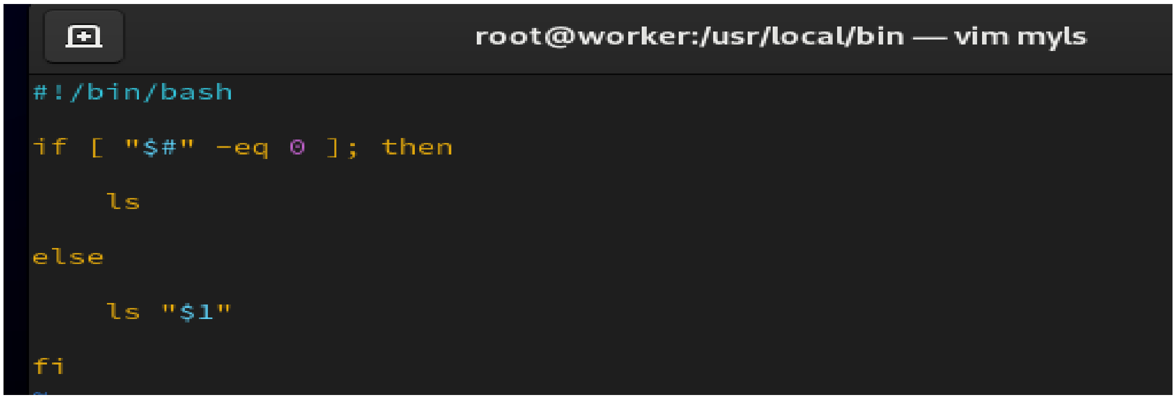
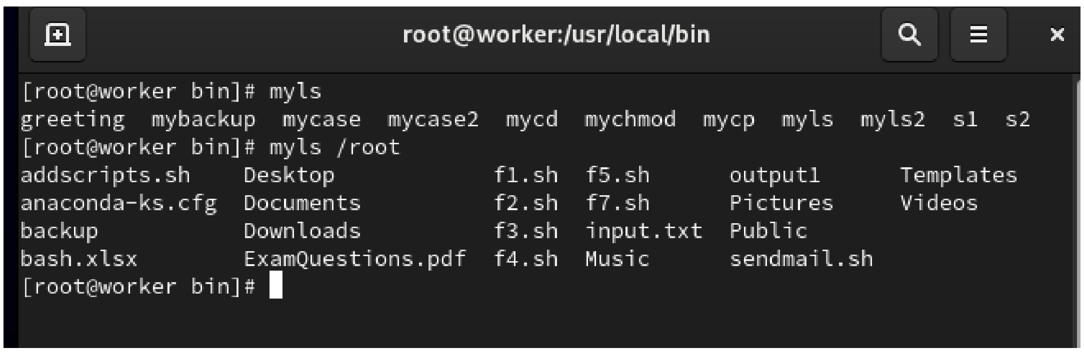
Answer:



root@worker:/usr/local/bin — vim myls

```bash
#!/bin/bash
if [ "$#" -eq 0 ]; then
    ls
else
    ls "$1"
fi
```

Verification:



root@worker:/usr/local/bin

```
[root@worker bin]# myls
greeting  mybackup  mycase  mycase2  mycd  mychmod  mycp  myls  myls2  s1  s2
[root@worker bin]# myls /root
addscripts.sh     Desktop            f1.sh  f5.sh      output1     Templates
anaconda-ks.cfg   Documents          f2.sh  f7.sh      Pictures    Videos
backup            Downloads          f3.sh  input.txt  Public
bash.xlsx         ExamQuestions.pdf  f4.sh  Music      sendmail.sh
[root@worker bin]# 
```

Q.6 Enhance the above script to support the following options individually:

a. –l: list in long format

b. –a: list all entries including the hiding files.

c. –d: if an argument is a directory, list only its name

d. –i: print inode number

e. –R: recursively list subdirectories

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Answer:

root@worker:/usr/local/bin — vim myls2

```bash
#!/bin/bash

if [ $# -eq 0 ]; then
    ls
else
    args="l a d i R"
    found=0
    for opt in $args; do
        if [ "$1" = "-$opt" ]; then
            ls -$opt $2
            found=1
        fi
    done
    if [ $found -eq 0 ]; then
        ls $1
    fi
fi
```

Verification:

root@worker:/usr/local/bin

```
[root@worker bin]# myls -l
total 44
-rwxr-xr-x. 1 root root 137 Apr  7 00:51 greeting
-rwxr-xr-x. 1 root root 141 Apr  8 03:04 mybackup
-rwxr-xr-x. 1 root root 337 Apr  8 00:57 mycase
-rwxr-xr-x. 1 root root 293 Apr  8 00:54 mycase2
-rwxr-xr-x. 1 root root 127 Apr  7 02:22 mycd
-rwxr-xr-x. 1 root root  48 Apr  8 02:46 mychmod
-rwxr-xr-x. 1 root root 110 Apr  8 04:04 mycp
-rwxr-xr-x. 1 root root  84 Apr  8 04:20 myls
-rwxr-xr-x. 1 root root 259 Apr  8 04:22 myls2
-rwxr-xr-x. 1 root root  69 Apr  7 02:02 s1
-rwxr-xr-x. 1 root root 142 Apr  7 02:02 s2
[root@worker bin]# myls -a
.   greeting  mycase   mycd     mycp  myls2  s2
..  mybackup  mycase2  mychmod  myls  s1
[root@worker bin]# myls -d
.
[root@worker bin]# myls -i
36171845 greeting  36170906 mycase2  36170920 mycp   36171848 s1
36170918 mybackup  36171858 mycd     36170929 myls   36171840 s2
36170903 mycase    36170917 mychmod  36170928 myls2
[root@worker bin]# myls -R
.:
greeting  mybackup  mycase  mycase2  mycd  mychmod  mycp  myls  myls2  s1  s2
[root@worker bin]# 
```

Bash Script Lab 1
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Questions on Lab1:

Q1. write addusers script ./addusers suffix count outfile.txt

Answer:

root@worker:/usr/local/bin — vim addusers

```bash
suffix=$1
count=$2
outfile=$3
for i in $(seq 1 $count)
do
    username="${suffix}${i}"

    #######check if user exist############
    id $username &>/dev/null
    while [ $? -eq 0 ]
    do
        username="${suffix}${RANDOM}"
        id $username &>/dev/null
    done

    useradd $username
    pass=$(openssl rand -base64 8)
    echo $pass | passwd --stdin $username > /dev/null
    chage -d 0 $username
    echo "$username,$pass" >> $outfile
    echo "User created: $username with password: $pass"
done
echo "All users created! Check $outfile"
```

Verification:

root@worker:/usr/local/bin

```
[root@worker bin]# addusers jimmy 3 output.txt
User created: jimmy1 with password: azovWVGtDi8=
User created: jimmy2 with password: YFdZ55c+vv8=
User created: jimmy3 with password: pZ2+EV4QS7M=
All users created! Check output.txt
[root@worker bin]# cat output.txt
jimmy1,azovWVGtDi8=
jimmy2,YFdZ55c+vv8=
jimmy3,pZ2+EV4QS7M=
[root@worker bin]# 
```

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Q1. write addusers script ./addusers suffix start end outfile.txt

Answer:

root@worker:/usr/local/bin — vim addusers2

```bash
#!/bin/bash
suffix=$1
start=$2
end=$3
outfile=$4
for i in $(seq $start $end)
do
    username="${suffix}${i}"

    #######check if user exist############
    id $username &>/dev/null
    while [ $? -eq 0 ]
    do
        username="${suffix}${RANDOM}"
        id $username &>/dev/null
    done
    useradd $username
    pass=$(openssl rand -base64 8)
    echo $pass | passwd --stdin $username > /dev/null
    chage -d 0 $username
    echo "$username,$pass" >> $outfile
    echo "User created: $username with password: $pass"
done
echo "All users created! Check $outfile"
```

Verification:

root@worker:/usr/local/bin

```
[root@worker bin]# addusers2 jimmy 4 6 output2.txt
User created: jimmy4 with password: VWIOKNT9Ooo=
User created: jimmy5 with password: djMUeeziPLg=
User created: jimmy6 with password: ma5oNyREfNM=
All users created! Check output2.txt
[root@worker bin]# cat output2.txt
jimmy4,VWIOKNT9Ooo=
jimmy5,djMUeeziPLg=
jimmy6,ma5oNyREfNM=
[root@worker bin]#
```

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Q3. write addusers script ./addusersfromfile data.csv outfile.txt (username ,password ,email)

Answer:

root@worker:/usr/local/bin — vim addusers3

```bash
#!/bin/bash
outfile=$2
inputfile=$1
while read myline
do
    fname=$(echo "$myline" | cut -d"," -f1)
    email=$(echo "$myline" | cut -d"," -f3)
    username=$fname
    ########check if user exist############
    id $username &>/dev/null
    while [ $? -eq 0 ]
    do
        username="${fname}${RANDOM}"
        id $username &>/dev/null
    done
    #######################################
    useradd -c "$email" $username    # -c adds comment field (field 5) in /etc/passwd
    pass=$(openssl rand -base64 10)
    echo $pass | passwd --stdin $username > /dev/null
    chage -d 0 $username
    echo "$username,$pass,$email" >> $outfile
    echo "User created: $username with email: $email"
done < $inputfile
echo "All users created! Check $outfile"
~
```

Verifictation:

root@worker:~

```
[root@worker ~]# cat input.txt
ahmed123,password123,darklizard1234@gmail.com
mohamed123,password1234,darklizard45@gmail.com
ali123,password12345,agml2047@gmail.com

[root@worker ~]# addusers3 input.txt output3.txt
User created: ahmed123 with email: darklizard1234@gmail.com
User created: mohamed123 with email: darklizard45@gmail.com
User created: ali123 with email: agml2047@gmail.com
useradd: invalid user name '19649': use --badname to ignore
passwd: Unknown user name '19649'.
chage: user '19649' does not exist in /etc/passwd
User created: 19649 with email:
All users created! Check output3.txt
[root@worker ~]# id ahmed123
uid=1012(ahmed123) gid=1012(ahmed123) groups=1012(ahmed123)
[root@worker ~]# id mohamed123
uid=1013(mohamed123) gid=1013(mohamed123) groups=1013(mohamed123)
[root@worker ~]# id ali123
uid=1014(ali123) gid=1014(ali123) groups=1014(ali123)
[root@worker ~]#
```

Bash Script Lab 1
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Q3. write grade script with if statement

Answer:

root@worker:/usr/local/bin — vim IFgrades

```
#!/bin/bash
if [ "$#" -ne 1 ]; then
    echo "Error: Missing input file."
    echo "Usage: ./grades.sh <input_file.txt>"
    exit 1
fi
input_file="$1"
if [ ! -f "$input_file" ]; then
    echo "Error: File '$input_file' not found."
    exit 1
fi
while read myline; do
    name=$(echo "$myline" | cut -d"," -f1)
    score=$(echo "$myline" | cut -d"," -f2)
    if [ "$score" -lt 50 ]; then
        grade="F (Fail)"
    elif [ "$score" -lt 65 ]; then
        grade="D (Pass)"
    elif [ "$score" -lt 75 ]; then
        grade="C (Good)"
    elif [ "$score" -lt 85 ]; then
        grade="B (Very Good)"
    elif [ "$score" -le 100 ]; then
        grade="A (Excellent)"
    else
        grade="Invalid Score"
    fi
        echo "Student: $name | Score: $score | Grade: $grade"
done < "$input_file"
-- INSERT --                                        4,9          Top
```

Q4. write grade script with case

Answer:

root@worker:/usr/local/bin — vim CASEgrades

```
#!/bin/bash
if [ "$#" -ne 1 ]; then
    echo "Error: Missing input file."
    echo "Usage: ./grades.sh <input_file.txt>"
    exit 1
fi
input_file="$1"
if [ ! -f "$input_file" ]; then
    echo "Error: File '$input_file' not found."
    exit 1
fi
while read myline; do
    name=$(echo "$myline" | cut -d"," -f1)
    score=$(echo "$myline" | cut -d"," -f2)
    case "$score" in
        100 | 9[0-9] | 8[5-9]) grade="A (Excellent)";;
        8[0-4] | 7[5-9]) grade="B (Very Good)" ;;
        7[0-4] | 6[5-9]) grade="C (Good)" ;;
        6[0-4] | 5[0-9]) grade="D (Pass)" ;;
        [0-9] | [1-4][0-9]) grade="F (Fail)" ;;
        *) grade="Invalid Score" ;;
    esac
    echo "Student: $name | Score: $score | Grade: $grade"
done < "$input_file"
~
```

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Q3,Q4 Verification:

root@worker:/usr/local/bin

```
[root@worker bin]# IFgrades grades.txt
Student: Ahmed | Score: 45 | Grade: F (Fail)
Student: Gamal | Score: 55 | Grade: D (Pass)
Student: Mohamed | Score: 70 | Grade: C (Good)
Student: Ali | Score: 82 | Grade: B (Very Good)
Student: jimmy | Score: 95 | Grade: A (Excellent)
[root@worker bin]# CASEgrades grades.txt
Student: Ahmed | Score: 45 | Grade: F (Fail)
Student: Gamal | Score: 55 | Grade: D (Pass)
Student: Mohamed | Score: 70 | Grade: C (Good)
Student: Ali | Score: 82 | Grade: B (Very Good)
Student: jimmy | Score: 95 | Grade: A (Excellent)
[root@worker bin]#
```

Q5. sendmail.sh script ./sendmail outfile.txt logfile

Answer:

root@worker:/usr/local/bin — vim sendmail

```
#!/bin/bash
input_file=$1
sender=oreog00039@gmail.com
token_password=kniwlslzpcmwwfkk
while read line
do
username=$(echo $line | cut -d "," -f1)
password=$(echo $line | cut -d "," -f2)
receiver=$(echo $line | cut -d "," -f3)
sub="ITI cloud Account"
fname=$(echo $username | tr -d [0-9])
mes="Dear $fname\n Your username is $username \n your password is $password. \n
 This email was sent to you by $sender "

# curl command for accessing the smtp server
curl -s --url 'smtps://smtp.gmail.com:465' --ssl-reqd \
--mail-from $sender \
--mail-rcpt $receiver \
--user $sender:$token_password \
-T <(echo -e "From: ${sender}\nTo: ${receiver}\nSubject: ${sub}\n\n${mes}")

done < "$input_file"
~
```

Verification:

1–3 of 3

To: agml2047	ITI cloud Account - Dear ali Your username is ali123 your password is password12345. This email was sent to you by oreog00039@gmail.com	7 Apr
To: darklizard45	ITI cloud Account - Dear mohamed Your username is mohamed123 your password is password1234. This email was sent to you by oreog00039@gmail.com	7 Apr
To: darklizard1.	ITI cloud Account - Dear ahmed Your username is ahmed123 your password is password123. This email was sent to you by oreog00039@gmail....	